

Combined virtual reality and haptic robotics induce space and movement invariant sensorimotor adaptation

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ABSTRACT

Prism adaptation is a method for studying visuomotor plasticity in healthy individuals, as well as for rehabilitating patients suffering spatial neglect. We developed a new set-up based on virtual-reality (VR) and haptic-robotics allowing us to induce sensorimotor adaptation and to reproduce the effect of prism adaptation in a more ecologically valid, yet experimentally controlled context. Participants were exposed to an immersive VR environment while controlling a virtual hand via a robotic-haptic device to reach virtual objects. During training, a rotational shift was induced between the position of the participant's real hand and that of the virtual hand in order to trigger sensorimotor recalibration. The use of VR and haptic-robotics allowed us to simulate and test multiple components of sensorimotor adaptation: training either peripersonal or extrapersonal space and testing generalization for the non-trained sector of space, and using active versus robot-guided reaching movements. Results from 60 neurologically intact participants show that participants exposed to the virtual shift were able to quickly adapt their reaching movements to aim correctly at the target objects. When the shift was removed, participants showed a systematic deviation of their movements during open-loop tasks in the direction opposite to that of the shift, which generalized to un-trained portions of space and occurred also when their movements were robotically-guided during the adaptation. Interestingly, follow-up questionnaires revealed that when the adaptation training was robotically-guided, participants were largely unaware of the mismatch between their hand and the virtual hand's position. The stability of the aftereffects, despite the changing experimental parameters, suggests that the induced sensory-motor adaptation does not rely on low-level processing of sensory stimuli during the training, but taps into high-level representations of space. Importantly, the flexibility of the trained space and the option of robotically-guided movements open novel possibilities of fine-tuning the training to patients' level of spatial and motor impairment, thus possibly resulting in a better outcome.

1. Introduction

Prism adaptation (PA) is a method nowadays extensively adopted in the context of rehabilitation of hemispatial neglect patients (Barrett et al., 2012; Champod et al., 2018). The original manipulation has been initially used to study visuo-motor plasticity in healthy participants, by introducing a deviation of visual reference frames and testing its effect on motor responses (Harris, 1963; Redding and Wallace, 1996). Conventionally, optical prisms are worn to laterally displace visual input by a fixed amount of degrees (usually 10°/15°), while participants are engaged in a pointing task towards visual targets. The displacement

causes a lateral error between the proprioceptively-guided pointing movements and the visual target. After perceiving this error, participants perform rapid recalibration of their motor responses, via both strategic and automatic mechanisms, in order to reduce the error and thus to correctly hit the target after few trials. In addition, a slower process of realignment takes place after a significant amount of trials (usually around 100) to consolidate the adaptation and bring the original and the shifted coordinate systems into correspondence (Harris, 1963; Bedford, 1999; Redding et al., 2005; Prablanc et al., 2020). The result of this correction is typically assessed by open-loop pointing tasks, performed after the removal of the prismatic lenses, in the absence of

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visual feedback from the pointing hand ('PA aftereffect'). The aftereffect of PA manifests as a pointing error in the direction opposite to the optical deviation induced by prism. In neglect patients, clinical studies report a PA aftereffect that spans beyond the motor domain: an amelioration was reported in perceptual and attentional tests for neglect assessment, such as line bisection (where participants have to mark the middle point of a line), exploration tasks, reading and functional abilities in everyday life. However, in healthy participants, there is conflicting evidence regarding the generalization of PA aftereffects to perceptual tasks: only left-deviating, but not right-deviating prism, were reported to induce a perceptual over-representation of the space in the direction opposite the prism in the line bisection task (see meta-analysis by McIntosh et al., 2019). Nevertheless, PA paradigm is a manipulation that is simple to implement in order to quickly modulate space representations (Colent et al., 2000; Michel et al., 2003).

Currently, PA training is provided via optical prisms during simple and repetitive reaching movements. Virtual reality might be an interesting innovation to implement a similar visuo-motor adaptation mechanism as PA, while adding multiple advantages: First, it allows a more flexible, fully controlled and automatized implementation of the adaptation than physical prisms. Second, it allows implementing the training in virtual environment reproducing multiple real-life scenarios and ecological tasks. Third, it allows full gamification of the training, including real-time feedback, which can potentially increase motivation. Finally, it allows training not only within the reaching space, but also in different sectors of space. However, the use of VR for neglect rehabilitation has been only sparsely reported (Katz et al., 2005; see Tsirlin et al., 2009 for review; Kim et al., 2011) and there are only few studies introducing an immersive VR version of PA in healthy participants (Carter et al., 2016; Gammeri et al., 2018; Ramos et al., 2019).

Furthermore, in neglect patients, PA is performed by using the healthy, ipsilesional hand, as most neglect patients are also paretic or even plegic in the contralesional hand. Robotics has been extensively used to mobilize the plegic upper limb in patients (Varalta et al., 2014; Choi et al., 2016), as well as to study the effect of visuo-motor adaptation in healthy participants (e.g., in 'force field adaptation' Kluzik et al., 2008 or in 'visuo-motor rotation' Cressman and Henriques 2009) in healthy participants, but never in the context of neglect rehabilitation and prism adaptation. In principle, using the contralesional, impaired limb during PA might boost the effect of the treatment by directly stimulating the neglected side of the body and the lesioned hemisphere. Moreover, the use of haptic-robotics can enhance learning, especially in patients with visual deficits, by providing multisensory feedback incorporating vision, sound, and touch.

In the present paper, we combined VR and haptic-robotics to implement a novel immersive virtual reality prism adaptation ('VR-PA') setup based on visuo-motor rotation. Participants performed reaching movements towards a virtual target, while a lateral displacement between the real position of their hand (signaled by proprioception) and the position of the virtual hand in the VR scenario (signaled by vision) was introduced (visuo-proprioceptive shift). The new haptic-robotic system used in the present study allowed us to compare adaption either while subject actively operated a robotic arm to reach targets or instead had their movements guided by the robot. In a first experiment, we tested possible motor and perceptual aftereffects, as those usually obtained by PA, with this novel haptic-robotics VR PA. In addition, we tested whether and how such effects generalize across different sectors of space, namely peripersonal and extrapersonal space. Thus, participants performed the adaption training either in the near or in the far space and motor and perceptual aftereffects were tested for both sectors of space. Their results were compared to those of another group of participants, who performed sham training, i.e., the same tasks with no visual-proprioceptive shift. In a second experiment, we tested the different contribution of active vs. passive movement. We compared a condition in which participants actively moved their arm while the robot was used only for providing haptic feedback (as in Experiment 1),

with a condition where the robot was used for guiding the participants' movement reproducing the kinematics recorded from a single participant during an active adaptation session. To assess the contribution of active versus passive motor control for virtual hand embodiment, we also tested subjective experience associated to the robotically guided adaptation, i.e., manipulation awareness, sense of agency and ownership.

The ultimate goal of this paper is to tease apart different components of the adaptation process – active versus passive movements and the associated awareness experience, action in the near versus far space – within a single experimental setup. These data will represent a basis to then implement a flexible and effective adaptation training via VR and robotics for neglect patients.

2. Methods

2.1. Methods overview

We conducted two visuomotor adaptation experiments. In Experiment 1, participants performed active reaching movements to targets in the near and far space, either with a 25° rightward shift between real and virtual hand (experimental group) or with no shift (sham group). In Experiment 2, adaptation was performed by robotically-guided reaching movements, without motor control, in the near space only, followed by an embodiment questionnaire testing participants' awareness of the manipulation. We measured adaptation aftereffects in the visuo-motor domain by means of open-loop reaching task, in motor-proprioceptive domain by a reaching straight ahead task, and in visuo-perceptual domain by means of the landmark task (details specified below).

2.2. Participants

A total of 60 healthy participants took part in the study (Mean $\pm$ SD age 25 ± 4 ; 32 males; 3 left-handed). All had normal or corrected to normal vision, and no neurological pathologies. All signed written informed consent, and the procedures were approved by the local ethical committee (CER-VD protocol no. 2017-01588). In experiment 1 (active motor execution): 32 participants took part in the VR-PA training session (16 trained in the near space, and 16 in the far space), and 21 participants took part in the sham session (10 in the near space, and 11 in the far space; one participant from the far group was not able to perform the experimental tasks and was therefore excluded from the analysis). There were 9 participants who took part in both the VR-PA training session and in the sham session (5 of them performed the VR-PA training session first, and 4 of them performed the sham session first). An independent subset of 16 participants took part in Experiment 2, the robot-guided VR-PA in the near space.

2.3. Experimental setup

2.3.1. Robotic haptic device

Participants used an arm-supporting haptic device that is able to stimulate proprioception in virtual reality (Fig. 1a; Mindmaze, patent number EP3177375A1). They rested their right arm on a robotic arm-support that moves freely in the horizontal plane (x and z axes), while grasping a wooden handle and thus moving the robotic arm in space. The range of motion using the device is 120 cm on the x axis (lateral movements) and 38 cm on the z axis (depth movement). The height was adjusted per participant to achieve maximum comfort, and remained constant throughout the experiment. For technical reasons, only the majority of participants (50) used an adjustable stool during the experiment, while the remaining participants (10) performed the experiment standing up. Using the device, they controlled the movements of a virtual hand and were able to receive naturalistic haptic feedback from interaction between this hand and objects in the virtual world. Haptic feedback was applied as forces transmitted by the robotic

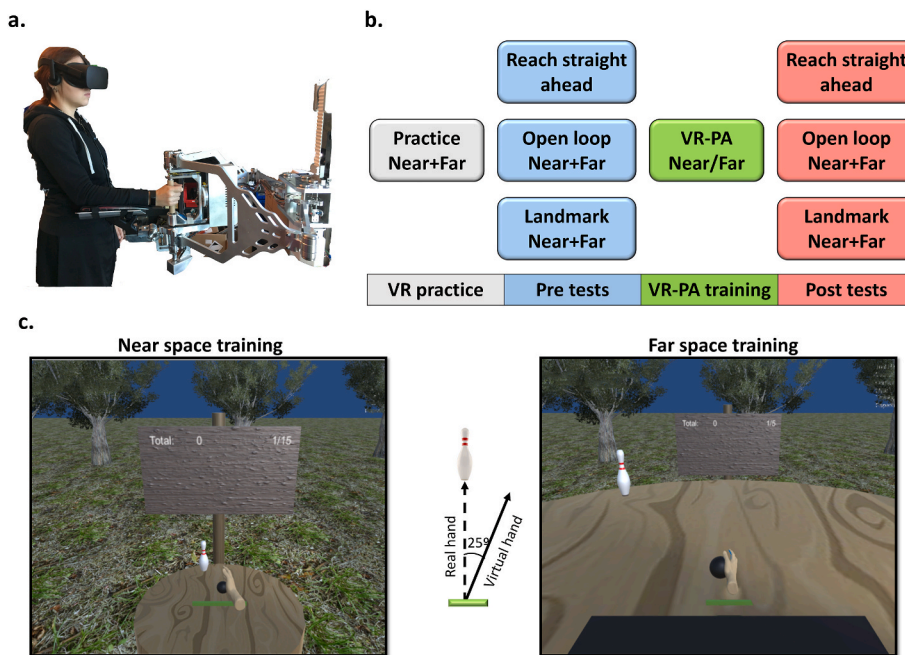


Fig. 1. Experimental paradigm and design. (a) The arm-supporting haptic device used by participants to move their hand in the virtual reality environment. (b) Sequence of experiments during a complete session. Note that the same set of four tests appeared before and after the VR-PA training (pre- and post-tests). For the sham group, there was sham-training instead of VR-PA training, in which no rotation shift was introduced. For the robot-guided group, the VR-PA was performed only in the near space, and movements were facilitated by the robot while subjects lay the arm passively on the arm-support (c) (the experimental setting during the VR-PA training. Each trial, participants had to reach and knock down the bowling pin at the edge of the table. The edge of the table could be either within reach (*left panel*), or out of reach with a bigger table radius (*right panel*). At one point, a clockwise rotational shift was introduced between the real hand movements and the virtual hand movements, to simulate the effect of optical prism (*middle*).

arm. In the guided-VR-PA condition, the robotic arm moved autonomously to guide the participants' reaching movement, while participant's arm rested passively on the arm-support and their hand grasped the wooden handle. The device is fully coupled with the virtual reality environment, so there is only negligible imperceptible delay between the participant's actual movements and those of the virtual hand. Furthermore, movement kinematics are fully tracked and are transferred online to the experimental computer. Finally, the grasping handle in the robotic arm includes a built-in trigger button whose presses are transmitted to the experimental computer in real-time.

2.3.2. Virtual reality environment

Participants wore a head-mounted display ('Oculus Rift', consumer version 1), and were immersed in a 3D virtual reality environment, which was created using Unity® software (version 2017.1.f1). The framerate was 500–1000 fps for the experimental computer, and a steady 120 fps for the subject view.

2.4. Experimental session procedures

2.4.1. Session overview – experiment 1 VR-PA condition

An experimental session lasted between 30 and 45 min. The sequence of experiments was as follows (see Fig. 1b): At the beginning of the session, participants performed a VR practice in order to get acquainted with the main task and setup. During this practice, they performed the VR practice task (described in section 2.4.1), i.e. reaching and knocking down a bowling pin with the virtual hand and ball with no shift. They performed 20 reaching trials in the near space, followed by 20 reach and roll trials in the far space. If by the last trial participants still had difficulty performing the task, they received an additional batch of 25 practice trials. After the last practice trial, the environment became completely dark so that the participant did not receive any visual cue. During this time, participants performed one reaching straight ahead movement (section 2.4.4).

After the practice, participants performed four baseline tests to establish their spatial lateral biases before adaptation ('pre-tests'). These tests included Open-loop experiments in the near and far space (see section 2.4.2), and Landmark experiments in the near and far space (see section 2.4.3). The order of experiments was counterbalanced between participants. After the pre-tests, participants performed the VR-PA

training either in the near or in the far space (see section 2.4.1).

Immediately following the VR-PA experiment (or the sham VR-PA), participants performed an additional reaching straight ahead in complete darkness. For some of the subjects in the VR-PA condition reaching straight ahead was not performed (6 out of 16 from the near train group and 7 out of 16 from the far train group).

Following this, participants repeated the open-loop and landmark tests (in near and far space) to determine their spatial lateral biases after the adaptation ('post-tests'). In the post-tests the open-loop tasks always appeared before the landmark tests.

2.4.2. Session overview – experiment 1 sham condition

The procedure for the sham group was identical to that of the VR-PA group (including identical pre- and post-tests), with the exception of participants performing a sham VR-PA training with no rotational shift. The training was performed (as in the VR-PA group) in either the near or far space.

2.4.3. Session overview – experiment 2 VR-PA robot-guided condition

Participants in the guided-VR-PA group performed training and aftereffect tests only in the near space. They started with active VR practice in the near space followed by reaching straight ahead and open-loop near test similar to Experiment 1 (Fig. 1b). However, they underwent the adaptation training passively, with the robot guiding their hand movements. The guided VR-PA was followed by active post-adaptation tests of reaching straight ahead and open-loop, similar to those of the other groups (for technical reasons, 3 of the subjects were not able to perform reaching straight ahead).

In the robot-guided VR-PA training session, the reaching movements towards the virtual targets were performed and guided exclusively by the robotic device. Participants were instructed to "pay attention to the target and think about moving your hand towards it, but let the robot do the movement for you". During the guided VR-PA training, 75 reaching movements were performed by the robot, replicating hand kinematics and target locations from a VR-PA session of a single subject in Experiment 1 (the active VR-PA group). The replay of actual biological reaching movement was chosen to induce a more naturalistic sensation in the guided condition. Each reaching movement was initiated by a trigger press made actively by the subject (as in the other conditions). As in the active VR-PA condition, the shift was introduced only at the 11th

trial. During the open-loop and the reaching straight ahead tests the subjects actively moved their hand as in the other conditions.

Following the VR session, participants from the robot-guided group had to fill out an embodiment questionnaire (section 2.4.5).

2.5. Virtual reality tasks stimuli and description

2.5.1. Virtual reality prism adaptation (VR-PA) training

Experimental setup: Participants used their right arm to move a natural-looking virtual hand grabbing a black bowling ball. In each trial participants had to reach and try to knock down a target (bowling pin, radius 5° visual angle) that appeared at the edge of a rounded table (Fig. 1c and supplementary videos <https://doi.org/10.1016/j.neuropsychologia.2020.107692>). The edge of the table was either within reach (table radius 30 cm; Fig. 1c, left - 'Near condition'), or out of reach (table radius 2 m; Fig. 1c, right - 'Far condition').

In the near condition participants could reach and touch the bowling pin, while the device provided naturalistic haptic feedback from it. The participant therefore had to push the bowling pin off the table while feeling a resistance from the object, simulating real-life conditions. Participants had to press the trigger button on the grasping handle while reaching, and release the trigger (while maintaining control of the ball) just before reaching the target. The presence of the ball in the near condition was meant to equalize visual inputs between the near and far conditions, such that any potential difference in results could not be attributed to low-level features of the sensory inputs.

In the far condition the bowling pin was out of reach, so participants had to reach and then roll the ball towards it. Reaching was performed by pressing and holding the trigger, and the ball rolled once the participant released the trigger. In the far condition there was no haptic feedback from the target. Target and ball sizes were adjusted so as to have the same visual angle in both the near and far conditions.

When participants managed to knock down the target, a rewarding signal involving sounds (magic-like bell chime) and vision (colorful lights around the knocked down target) was presented.

In both near and far conditions, the participants had to start their movement from a point of origin located just in front of the midline of their bodies, and had to return to this position at the end of each trial. In Unity this position corresponded to the coordinates (0, 0) on the XZ plane. The point of origin was marked by a thin line in the middle of the virtual table that turned green when the participant approached it (22 × 0.9 cm; Fig. 1c), and the participant's hand was gradually pulled to it at the end of each trial by forces exerted by the robotic device, to ensure that the movement kinematics in terms of origin and goal was properly defined and standardized across participants and trials.

The virtual environment was an outside scene, containing naturalistic features such as trees, grass, etc., to create an enjoyable, immersive experience. A scoreboard was placed behind the virtual table, which presented number of scored points after a successful hit, in order to provide feedback about the performance and enhance motivation during the training.

Procedure: During the VR-PA experiment only the ball was visible and represented the participants' hand position, while the virtual hand remained invisible in the virtual environment. The ball became visible only when participants reached for more than 10 cm away from the origin point, so no visual feedback about the hand position was given at the origin. This partial visual feedback of hand trajectory presents a compromise between concurrent and terminal shift, that will enable both strategic and online mechanisms to take place (Facchin et al., 2020). The ball remained visible until participants finished their reaching movement, and reward was given in case of a hit as in the practice trials. The next target was launched only once the participant's hand went back into the origin point. Targets were spawned in a radius of 30 cm for the Near condition and 2 m for the Far condition. Previous studies found that increasing the span of trained lateral positions improves generalization of adaptation aftereffects to new positions

(Krakauer et al., 2000). Therefore, in the current study targets had random lateral position in the range of -45° – 45° degrees of visual angle from the origin point. The training contained a total of 50 trials: participants performed 10 trials without any visual shift, and then at trial 10, a rotational shift of 25° to the right was induced between the actual hand trajectory and the virtual ball trajectory (Fig. 1c, middle). A shift of 25° to the right was selected because it presents a good balance between effective deviation with potential applications for neglect rehabilitation, and maintaining ownership for the virtual hand (Facchin et al., 2013; Fang et al., 2019). A larger number of trials is needed in some cases to reach true adaptation (Redding et al., 2005; Prablanc et al., 2020). Therefore, participants who were not able to obtain a successful performance by end of the 50 trials were given an extra batch of 25 trials (13 out of 16 participants in the far training; 7 out of 16 participants in the near training). Consequently, subjects in the near group performed between 50 and 100 VR-PA trials (median = 72.5), and subjects in the far group performed between 50 and 125 VR-PA trials (median = 100). The control group performed a sham VR-PA experiment, in which the procedure was identical, except that no rotational shift was induced.

In our setup, to get an immersive sensation, the orientation of the training environment coincided with the participant's own egocentric point of view. Correspondingly, we chose to use a rotation instead of a translation between the virtual hand and the real hand. This method was chosen to keep the shift angle constant, regardless of the distance from the origin position.

2.5.2. Open-loop task

Experimental setup: The open-loop experiment was similar to the VR-PA experiment, except that participants did not receive any visual or haptic feedback regarding their hand position. Namely, the virtual hand and ball were not visible at any point in the experiment. The participants began the experiment by being pulled to the origin point, and then, after the target appeared, they had to press the trigger and reach for the target based on proprioception alone. In the near condition, the target disappeared a moment before participants completed their reach movement. In the far condition, the target disappeared at the end of the reach movement when participants released the trigger. In both conditions, no feedback was given regarding the accuracy of the reach movement. Importantly, to diminish strategic re-aiming, the participants were informed that the post- and pre-training tests were similar, namely that there was no shift induced in the open-loop tests.

Procedure: Participants performed 20 trials in which they had to reach towards a target located at the edge of the table without receiving any sensory information about their hand position. The targets were spawned randomly in different lateral positions at the edge of the table. This experiment was performed before and after the VR-PA training, in both the near and the far space.

2.5.3. Landmark task

Experimental setup: In the landmark task, a wooden stand containing a shelf was located in front of the participants. A red can that served as a fixation point was placed in the middle of the shelf. In each trial, two bowling pins briefly appeared to the left and the right of the can (Fig. 5a). Participants' task was to determine whether the left or the right target was farther away from the red can. The stand and the shelf were located either within reach of the participants (30 cm distance; 'Near condition'), or out of their reach (2 m distance; 'Far condition'). The sizes of the shelf, the can and the bowling pins were adjusted to have the same visual angle in both the near and the far conditions. The participants' hand remained fixed in front of the midline of their body by forces exerted by the robotic device.

Procedure: Participants performed 50 trials of the landmark task. To start a trial, participants were told to fixate on the red can in the middle of the shelf and press the trigger button. Then the two bowling pins briefly appeared for a duration of 150 ms to the left and the right of the can, after which the participants had to orally respond "left" or "right" to

decide which pin is farther away from the can. The experimenter recorded their responses by pressing the left or right arrow buttons on a keyboard, in order to keep the participant free from any motor task. The difference in distances between the pins and the can (right pin distance – left pin distance) ranged from -2 to 2 cm in the far condition, and -0.6 to 0.6 cm in the near condition (minus values indicate cases where left target was farther away from the center), the number of left farther and right farther trials was the same. Importantly, in 12 out of 50 trials the targets were in equal distance from the center (similar to Makin et al., 2010; Schintu et al., 2014). The order of the trials within an experiment was pseudorandomized, but kept constant between the experiment repetitions in pre-tests and post-tests.

2.5.4. Reaching straight ahead task

One reaching straight ahead movement in darkness was performed immediately after the virtual reality practice, and another was performed immediately after the VR-PA training. The environment became completely dark so participants could not perceive any visual stimulus within their field of view. Participants had to perform a reaching movement straight ahead using the robotic device, and then bring their hand back to the origin point ('reaching straight ahead'). Their hand was again pulled back to the origin at the end of the movement. Participants received no visual or haptic feedback during this reaching task.

2.5.5. Embodiment questionnaire

Since the passive condition infers a loss of feeling of agency, we assessed agency and ownership following the training. At the end of the guided VR-PA session, subjects filled out a questionnaire assessing embodiment of the virtual hand during the training. The questionnaire included questions regarding the virtual and real hand co-location ("it seemed like my hand was in the location where the virtual hand was"), feeling of agency ("It seemed like I was in control of the virtual hand") and of ownership ("It seemed like the virtual hand was my hand"; (Bassolino et al., 2018). Participants rated their responses 0–6 (strongly disagree–strongly agree).

Since questionnaire data was not collected for Experiment 1, in order to measure manipulation awareness, agency and ownership during active PA, we used as comparison questionnaires from a dataset in which subjects performed a similar adaptation training in immersive VR, with the same amount of displacement. Identical questionnaire data was collected from two additional groups of subjects ($N = 26$, age 25 ± 2.8 years, 10 females) following a similar paradigm: participants had to reach targets in a VR environment, but using an oculus controller (a light remote control placed in the hand) instead of the robotic device. The first group performed 160 trials with 20° rotational shift between the virtual and real hand ('shift VR'), while the second group performed the same task without any shift ('no shift VR'). A 2-way ANOVA followed by post-hoc tests was performed on the questionnaire scores of the three conditions, with factors group $\times$ question type.

2.6. Data analysis and statistical tests

All data analysis was done using costumed scripts written in Matlab (Mathworks, Natick, MA, USA), with complementary statistical tests in R 3.4.3 and JASP 0.11.1.

2.6.1. Kinematic data analysis

Hand kinematics were recorded continuously throughout the experimental session by logging the position of the virtual hand in each frame. To obtain reach trajectory in each trial, data samples were separated into the different trials, then the trial data points were defined from the first sample point of each trial until the point of maximum reach (maximal distance from the origin). In order to visualize the different trial kinematics for each participant, trials were binned to left/middle/right according to target position. Trials in which targets appeared between -45° and -17° were classified as left, -17° till 17°

were classified as middle, and 17° till 45° were classified as right. The hand trajectories binned to each type were adjusted according to a single representative left/middle/right target position.

2.6.2. Target error analysis

The angular deviation of the reach movement from the center of the target was calculated in each trial. For the VR-PA and open loop conditions, the point of reference for the deviation calculation was taken as the position of the virtual hand immediately after it became visible (at 10 cm distance from the origin position). This point was taken and not the final reach position, in order to take into account only the ballistic component of the movement and not the online correction. Then target error in each trial was calculated as the difference between the reach trajectory angle and the target angle (reach angle – target angle), where negative values indicated that reaches were to the left of the target and positive values indicated that reaches were to the right of the target. Target errors were averaged across trials in each experiment for statistical analysis (section 2.4.4).

In the reaching straight ahead task the point of reference was taken as the point of maximal reach, and its angular deviation from zero was considered for statistics.

2.6.3. Landmark point of subjective equality analysis

For the landmark task, a psychometric curve was fitted for each participant and each experiment, according to the different target positions and the subjects' left/right responses (Fig. 5b). Then, the point of subjective equality (PSE) was taken as the point of the curve for which the subject had 50% chance of reporting 'right' and 50% chance of reporting 'left'. The PSE estimates at which configuration of the targets the participant would perceive the targets as being at equal distances from the center. A positive PSE means that the right target has to be farther away from the center in order for the distances to be perceived as equal, which implies an over-representation of the left space, and vice-versa for negative PSE and over representation of the right space. The PSE value of each experiment was used for statistical analysis.

2.6.4. Statistical tests

To assess difference between groups in the open-loop and landmark conditions, a 3-way mixed-effect repeated-measures analysis of variance (ANOVA) design was implemented using JASP software. The within-subject factors were test space (near/far) and test phase (pre/post), while the between-subject factor was VR-PA training space (near/far). Effect sizes were computed in JASP as Generalized Eta Squared (η^2 2G; Olejnik and Algina, 2003).

Additionally, a median-split was performed to assess the effect of the number of VR-PA trials on subsequent open-loop aftereffects. Participants in the VR-PA group were divided into two groups of more versus less trials during training above or below the median of their training group (median = 72.5 trials in the near group, and median = 100 trials in the far group; two subjects in the far group were exactly at the median and were therefore excluded from the analysis), and a 3-way ANOVA was used with test-phase and test-space as within subject variables, and trial number (more/less) as a between subjects variable. For the reaching straight ahead task, because there were different sample sizes in each experimental group (not all participants performed this task), a 3-way mixed effect ANOVA was used, with training space (near/far), group (experimental/sham) as between-subject factor, and test phase (pre-post) as within-subject factor. The main effects and interactions were tested using Kenward-Roger approximation for degrees of freedom as implemented by R package 'afex' and were FDR-corrected for multiple comparisons.

For the guided VR-PA group a paired *t*-test was performed between the open-loop results pre- and post-the adaptation (only near space was performed), and a 2-way repeated-measures ANOVA was performed when pooling together the group who performed adaptation in Experiment 1 in the near space to the robot-guided adaptation group.

3. Results

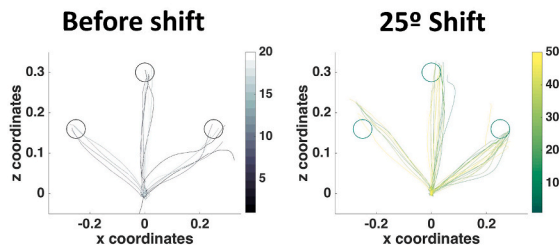
3.1. Experiment 1 – active VR-PA

3.1.1. VR-PA - adaptation training stage

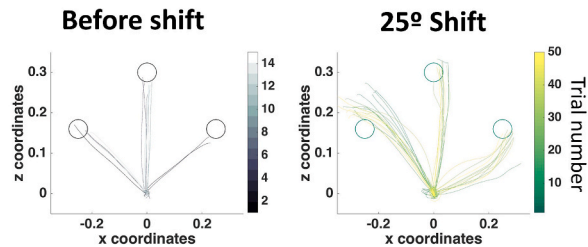
We first analyzed the average pointing error during the adaptation stage in the experimental groups (25° shift) and the sham groups (no

shift) training in near or far space. The average error showed a significant effect of group ($F(1,47) = 43.5$; $p < 0.001$; $\eta^2_G = 0.48$), but no effect of space or interaction between group and space ($p > 0.37$). As expected from the manipulation, participants from the experimental group made on average higher reaching errors, whereas no deviation from target was found in the sham group (mean deviation $4.02 \pm 0.37^\circ$ for the experimental group and $-0.09 \pm 0.47^\circ$ for the sham group). Fig. 2a and b

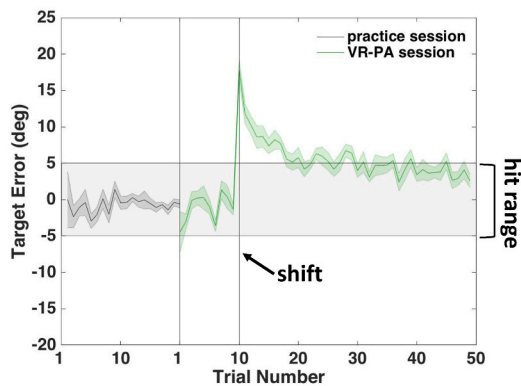
a. Near space kinematics



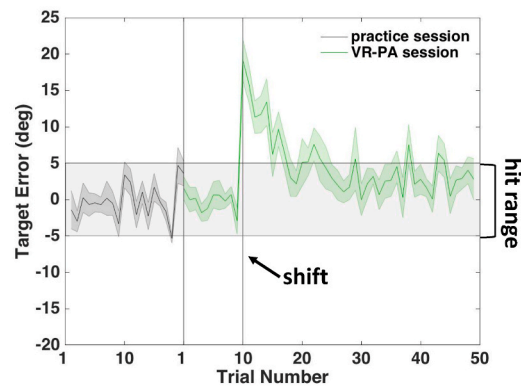
b. Far space kinematics



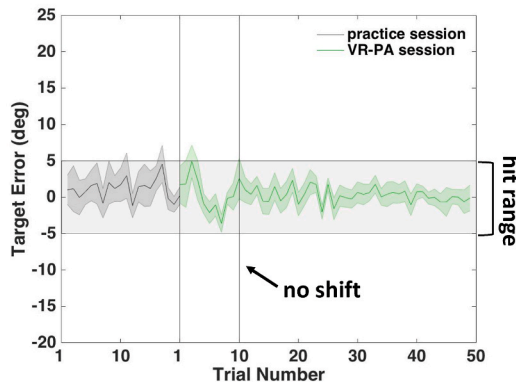
c. VR-PA training near space (n=16)



d. VR-PA training far space (n=16)



e. Sham training near space (n=10)



f. Sham training far space (n=10)

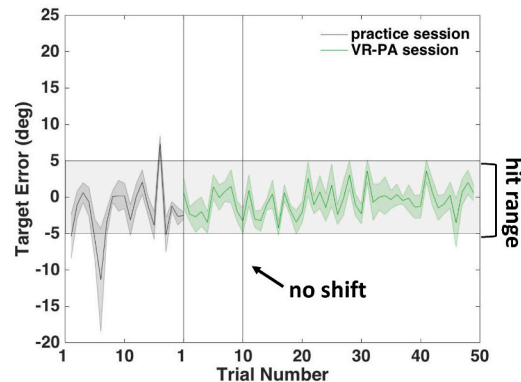


Fig. 2. Reaching kinematics and accuracy during VR-PA training experiment. (a–b) The movement kinematics in each trial was plotted for two representative participants: one training in the near space (a) and the other training in the far space (b). Targets were binned to left/middle/right locations to facilitate visual inspection (during the experiment the targets were spawned in random locations). The 3 circles represent the 3 target types (left/middle/right). The hand trajectories binned to each type were adjusted accordingly and then plotted together with the representative left/middle/right target. The far targets were projected to appear as if they were in near space to facilitate visual inspection of aiming. On the left panels is the motor behavior of the participants in the practice trial, i.e. before inducing the shift. On the right panel are hand kinematics after inducing a rotational shift of 25° to the right. The colorbar denotes the progression of trials across the experiment, whereas grayscale plots on the left are of the VR practice before a shift was introduced, and the green shades plots are of the actual VR-PA training with a shift. (c–f) The average deviation from the target for each trial during the VR practice (gray) and the VR-PA or sham-VR-PA training (green). Shaded contours denote the SEMs between subjects. Participants were trained either in the near space (c) or in the far space (d). The gray rectangle denotes the range of angles in which a participant will hit the target. The vertical lines represent the separation between the VR practice and the VR-PA training, and the timing of the shift for training groups (c–d). No shift was introduced for the sham groups in the near (e) or far (f). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

depict an example trial-by-trial hand kinematics of a participant trained in the near, and a participant trained in the far space, respectively. As can be seen, before inducing the shift participants aimed accurately at the targets in both near and far space (Fig. 2a and b; left panel). After inducing the shift, participants perform online corrections to reach the targets at first (Fig. 2a and b; right panel; dark green shades), but by the end of the session they recalibrate their initial aiming position to compensate for the shift (Fig. 2a and b; right panel; light green and yellow shades).

We then investigated adaptation dynamics in our four groups. Fig. 2c–f shows the trial by trial mean deviation from the target, averaged across participants. For the experimental groups, reaching errors emerged as soon as the shift was induced, and the error progressively decreased until participants were again able to successfully hit the target (movements fall within the gray target area). In order to quantify the changes in reaching error throughout the experiment, we divided the experiment into 5 epochs, each containing 10 trials, and conducted a $2 \times 2 \times 5$ repeated measures ANOVA with group (experimental/sham), space (near/far), and epoch (1–5) as factors. We found a significant effect of epoch ($F(4,188) = 17.12$; $p < 0.001$; $\eta^2_G = 0.183$), and an interaction between epoch and group ($F(4,188) = 18.62$; $p < 0.001$; $\eta^2_G = 0.196$). Next, we performed 2-way ANOVA on the experimental groups with space and epoch as factors. We found a significant effect of epoch ($F(4,120) = 38.68$; $p < 0.001$; $\eta^2_G = 0.459$), and no effect of space or interaction between space and epoch. Post hoc tests revealed that the 2nd epoch, in which the shift was introduced, was significantly different

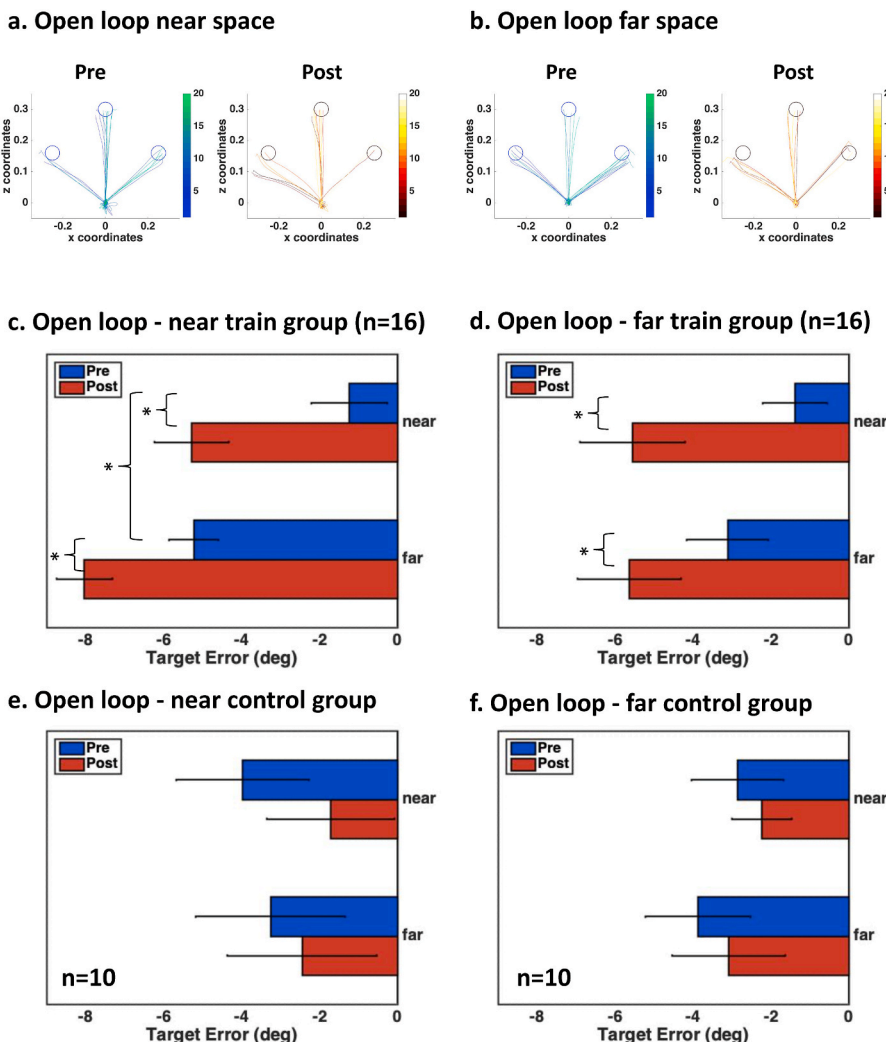
from all the other epochs ($p < 0.001$), while epochs 3–5 were similar. This implies that the majority of adaptation occurred within the first 10 trials of the shift onset.

Although the overall adaptation dynamics was similar between near and far space, the variability between trials looked different. Indeed, a 2-way ANOVA on the variance between trials with space and group as factors showed a main effect of space ($F(1,47) = 10.863$; $p = 0.002$; $\eta^2_p = 0.188$), and no interaction between group and space. Indeed, aiming accuracy was more variable in the far space than in the near space (mean variance $\pm$ SEM 29 ± 3.4 in the near and 72 ± 11.7 in the far space), regardless of the presence of a shift, probably due to reduced online correction possibility when the movement ends far from the target. Overall our results show that participants were equally able to adapt in the near and in the far space, with more variability between trials in the far space.

3.1.2. Open-loop reaching

Next, we assessed the training-induced lateral spatial bias (aftereffect) during active reaching with no visuo-tactile feedback of hand position, namely the open-loop task. The task was similar to the VR-PA training task, except that no feedback from the hand position was given. In line with classic PA literature, we hypothesized a systematic deviation of reaching movements in the absence of visual feedback in the direction opposite of the shift induced by VR-PA. To test this prediction, we compared pre-training and post-training reaching accuracy in the open-loop task. Fig. 3a and b presents the open-loop kinematics of an example

Fig. 3. Reaching kinematics and accuracy during Open-loop task pre- and post-training. (a–b) Hand kinematics of a representative participant during the open-loop task (no visual feedback from hand) in the near (a) and far (b) space, before (blue shades; left panels) and after (red shades; right panels) VR-PA training. Specifications are as in Fig. 2. Note that in contrast to the VR-PA training, the participants did not have the possibility to online-correct their movements, due to lack of feedback regarding hand position. Therefore, the reach trajectory is rather straight in all the trials of the open-loop task. (c–f) Average target error across trials and participants in the four types of open-loop tests each participant performed (near-far-pre-post). The errorbars denote SEMs between subjects, and asterisks denote significant differences between bars. Each plot represents a different group of participants: (c) trained in near VR-PA; (d) trained in far VR-PA; (e) sham group trained in near sham-VR-PA; (f) sham group trained in far sham-VR-PA. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



participant in the near and far space, before and after VR-PA training. As can be seen, even without visual feedback of the hand position, based on proprioception alone, the participant was quite accurate in reaching the targets, both in the near and in the far space before VR-PA. However, while performing the same task after VR-PA training, the participant showed a systematic bias towards the left (opposite to the VR-PA shift), namely a VR-PA aftereffect.

In order to demonstrate this effect statistically, we then averaged the target reaching error across trials in each of the experimental conditions to find the overall aftereffects in the open-loop task. We first present the results of the Experimental groups. Fig. 3c and d shows the open-loop results of the groups who performed VR-PA training in the near and far space, respectively. We performed a $2 \times 2 \times 2$ mixed ANOVA with the factors test-phase (pre and post) and test-space (near and far testing) as within subject factors, and train-space (near or far training) as a between subject factor. It revealed a significant main effect of test phase, namely a significant difference between pre- and post-training ($F(1,30) = 40.24$; $p < 0.001$; $\eta^2_G = 0.159$; average deviation angle of 6.1°). Train-space factor (near/far VR-PA training) did not have a significant effect on the open-loop results ($F(1,30) = 0.95$; $p > 0.3$), and there was no two-way interaction between phase and train-space ($p = 0.94$), implying that the overall pattern was similar between the groups who trained in different sectors of space. However, there was a significant main effect of test-space (near/far testing; $F(1,30) = 10.13$; $p = 0.003$; $\eta^2_G = 0.07$), and a trend towards interaction between train-space and test-space ($F(1,30) = 3.34$; $p = 0.07$; $\eta^2_G = 0.024$), deriving from the overall less accurate performance in the far space, especially for the near training group.

The same analyses were then performed on the sham groups, which showed relatively small errors in their reaching movements, with no difference between pre- and post-training performance (Fig. 3e and f) and no differences between test-spaces, as there was no significant main effect or interaction for the open-loop task in the sham group (all p -values > 0.19).

These results demonstrate that for the open-loop task an aftereffect is induced by the VR-PA training, both in the near and far space, following training in either sector of space.

Because some of our participants performed a different number of VR-PA trials (see methods, section 2.4.1), we wanted to validate that the number of adaptation trials did not directly affect the size of the aftereffect, namely that all of our subjects achieved true adaptation and subsequent aftereffects, despite any difference in the number of trials performed. To that end, an additional median-split analysis was performed to assess the effect of the number of VR-PA trials on subsequent open-loop aftereffects. A 3-way ANOVA (test-space $\times$ test-phase $\times$ trial-number), pooling together participants who trained in the near and in the far space, revealed no significant main effect, 2-way or 3-way interaction between the number of VR-PA trials (more/less trials) and the open-loop aftereffect. To see the results in details, we repeated the analysis while separating the near-trained and far-trained groups, as the two groups received on average a different number of trials. Consistently, there was no significant 2-way or 3-way interaction of trial number in either the near or the far group (see Figure S1). The main effect of test-phase was still evident even after reducing group sizes with the median split ($F(1,14) = 17.57$, $p < 0.001$ for the near group; $F(1,12) = 18.86$, $p < 0.001$ for the far group).

Our results suggest that in our participants, the absolute amount of adaptation trials did not directly affect the size of the open-loop aftereffects. This might imply that in any case a sufficient number of trials was performed for obtaining an adaptation aftereffect, within the range of inducing true realignment.

3.1.3. Reaching straight ahead

The previous analysis does not allow us to determine whether the VR-PA induced aftereffect is constrained to the specific experimental context in which the training was performed, or whether it represents a more general sensory-motor realignment independent of the context. To

approach this question, we tested the participants on a stimulus-free task in which they had to use the robotic arm to reach straight ahead in complete absence of visual feedback. This task is equivalent to pointing straight ahead task that is commonly used for detecting lateral spatial biases in the perception of the body midline both in neglect patients (e.g., Heilman et al., 1983) and in healthy participants after PA (e.g., Schintu et al., 2014). The participants performed this task once immediately after the VR practice (before any visual deviation), and a second time immediately after the VR-PA training. Reaching straight ahead kinematics are depicted in Fig. 4a and b for a representative subject who participated in both a VR-PA training session and a sham session. As can be seen, the reach movement was deviated to the left immediately after VR-PA training, but remained accurate after sham-VR-PA. The accuracy in reaching straight ahead, i.e., the deviation from 0° , was calculated for each subject. We conducted a $2 \times 2 \times 2$ mixed ANOVA with test-phase (pre-post) as within factor and trained-space (near/far training) and group (experimental/sham) as between factors. There was a main effect of test-phase ($F(1,35) = 10.02$, $p = 0.003$; $\eta^2_G = 0.119$) and a marginally significant 2-way interaction between group and test-phase ($F(1,35) = 3.09$, $p = 0.08$; $\eta^2_G = 0.04$). In order to explore the source of this possible interaction, we ran 2 separate ANOVAs for the experimental and sham groups; We found a significant main effect of test-phase in the VR-PA training group (Fig. 4c and d; $F(1,17) = 10$; $p = 0.006$; $\eta^2_G = 0.227$), but not in the sham group (Fig. 4e and f; $p = 0.28$). These results imply that the VR-PA induced a spatial deviation of sensory-motor behavior that extended beyond the context of the specific VR-PA experiment, generalizing also to an internally driven proprioceptive task without external visual cues (see Figure S1).

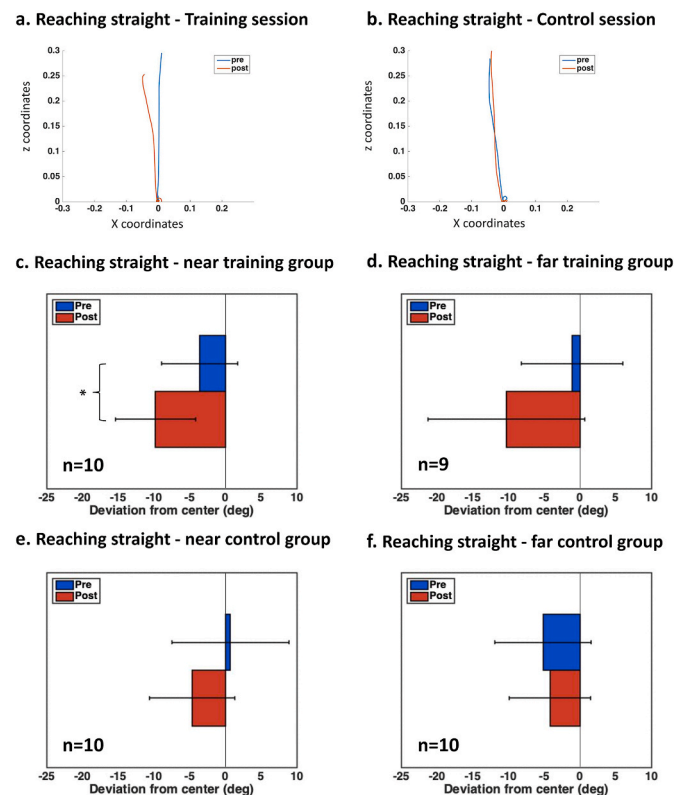


Fig. 4. Kinematics and accuracy of reaching straight ahead task. (a–b) Hand movement kinematics of a representative subject reaching straight ahead before and after (a) a real VR-PA training, and (b) a sham-VR-PA training (data were collected in different sessions). (c–f) Mean deviation from center in degrees before and after training: (c) VR-PA training in near space (d) VR-PA training in far space (e) sham-training in near space (f) sham-training in far space. Errorbars denote the SEMs between subjects.

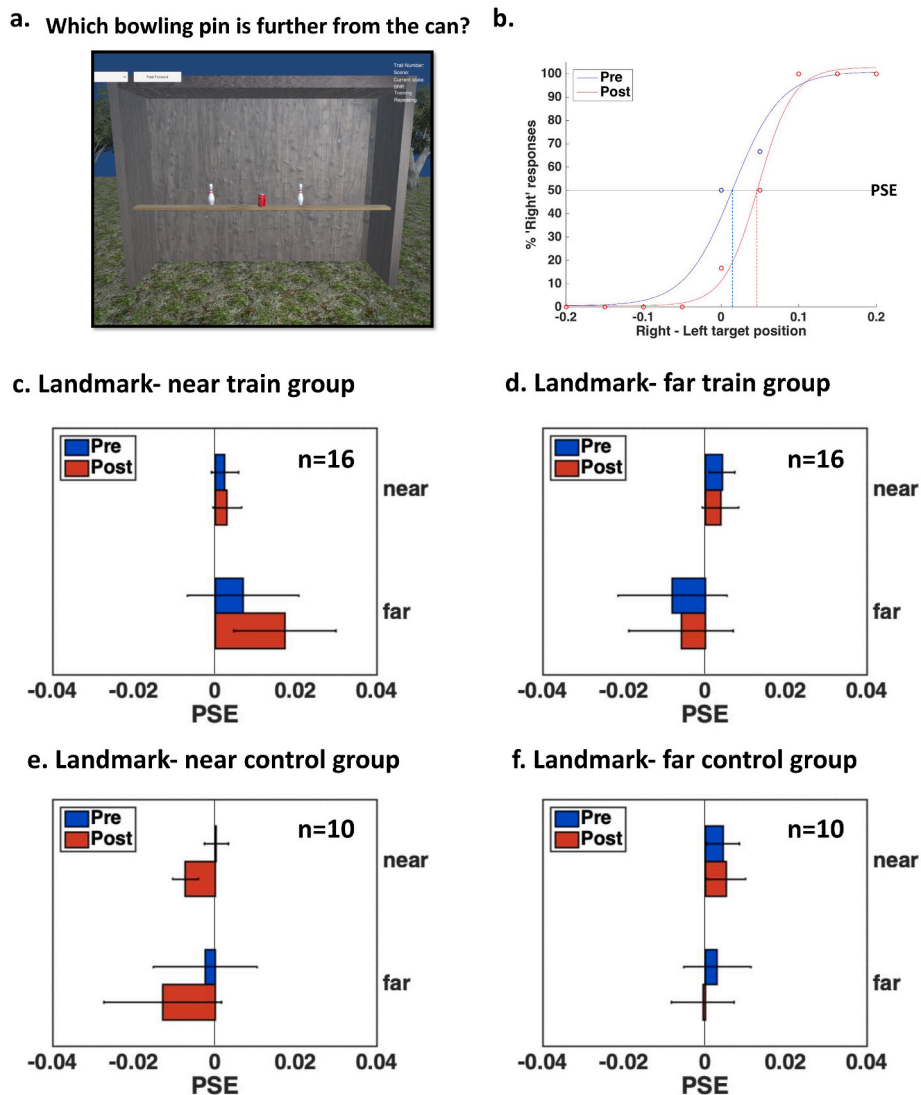


Fig. 5. Performance in the perceptual landmark task. (a) Experimental paradigm and stimuli for the VR landmark task. (b) An example of a psychometric curve from which pre- and post-training point of subjective equality (PSE) were extracted. (c–f) Mean PSE for each condition (near-far-pre-post) in the (c) group trained on VR-PA in near space (d) grouped trained on VR-PA in far space (e) sham-training in near space (f) sham-training in far space. Errorbars are SEMs between subjects.

3.1.4. Landmark

Next, we aimed to examine whether the measured aftereffect is confined to tasks requiring a motor response, or can also affect spatial perception when no movement is required. Therefore, we tested participants before and after VR-PA training with the landmark task, which detects biases in visuo-spatial perception without requiring hand movement. Participants had to determine which of two briefly presented targets was farther away from a central fixation target (Fig. 5a). They performed the task both in the near and in the far space. We calculated the PSE for each participant in each space (see e.g., Fig. 5b) and compared the average PSE pre- and post-training in the near and far space (Fig. 5c–5d). We performed a $2 \times 2 \times 2$ mixed ANOVA on the PSE values, with the factors train-space, test-space, and test-phase (pre-, post-), and found no main effect of test phase ($p = 0.38$) and no interaction between test-phase and test-space ($p = 0.2$). The results of the sham group also showed no spatial biases (Fig. 5e–5f; all $p > 0.29$). These findings reveal that the VR-PA training had no significant effect on biasing spatial perception in the landmark task.

Overall, the results of Experiment 1 suggest that VR-PA training created a motor aftereffect regardless of the training space or the experimental test setup, but that the effect did not transfer to a perceptual task not involving an active arm movement.

3.2. Experiment 2 - robot-guided VR-PA

In the robot-guided VR-PA condition, participants passively rested their hand on the robotic arm-support, while the robot replayed kinematics of one of the active VR-PA adaptation sessions to guide their reaching movements. Consequently, the kinematics of the adaptation phase in the robot-guided condition were identical for all the participants. Therefore, we report only the results of the aftereffects tests, which the participants actively performed before and after the robot-guided adaptation.

3.2.1. Open loop reaching - guided

Fig. 6a presents the open-loop kinematics of an example participant before and after robot-guided VR-PA training. As can be seen, reaching movements were accurate (as in the other experimental groups) prior to guided-VR-PA session, even without vision of the hand. Following rightward guided-VR-PA, reaching movements were recalibrated, so participants were aiming to the left of the targets. This occurred even though participants did not need to actively adapt the movement during the VR-PA session, but only to observe and feel the adaptation being performed by the robot. Overall, we found a significant difference between average reaching error before and after guided-VR-PA, i.e.,

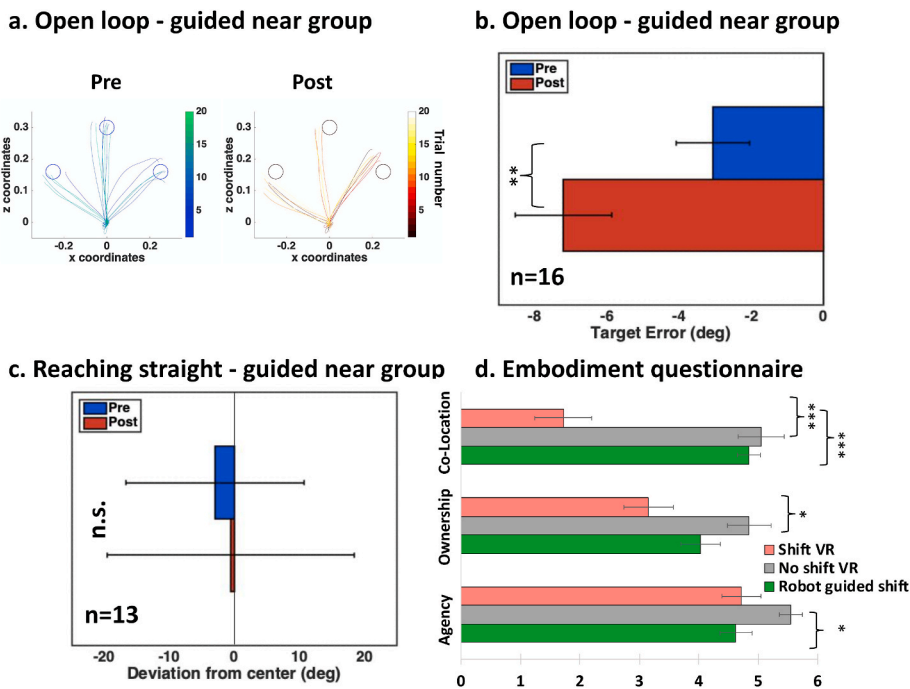


Fig. 6. Open-loop aftereffects were present also after robot-guided VR-PA. (a) Hand kinematics of a representative participant during the open loop task before (blue shades; left panels) and after (red shades; right panels) robot-guided VR-PA training. Specifications are as in Fig. 2. (b) Average target error across participants pre and post guided VR-PA. The error bars denote SEMs between subjects, $**p < 0.005$. (c) Average deviation from true midline in reaching straight ahead pre and post guided VR-PA. (d) Results of embodiment questionnaire in robot-guided VR-PA, active VR-PA, and active sham (no shift) VR training. $*p < 0.05$; $***p < 0.0005$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

movements were more biased to the left following the passive adaptation process (Fig. 6b; paired t -test, $p = 0.001$).

In order to test whether the same amount of motor aftereffect was found in the active and robot-guided adaptation conditions, we compared the results of the open loop task in the near group from Experiment 1 (trained and tested in the near space) with data from Experiment 2 (2x2 ANOVA). We found a significant main effect of test-phase ($F(1,30) = 26.98$, $p < 0.001$; $\eta^2_G = 0.193$), and no interaction ($F(1,30) = 0.003$, $p = 0.96$), implying similar and robust pre-post effect in the active and the robot-guided conditions.

3.2.2. Reaching straight ahead - guided

In the guided condition, reaching straight ahead results had large variability between participants and overall no significant difference was found between pre and post guided VR-PA ($t(15) = -0.57$, $p = 0.7$; Fig. 6c), unlike in the active adaptation condition.

3.2.3. Embodiment questionnaire

Since the passive training condition putatively implies a loss of feeling of agency over the hand movement, we introduced an embodiment questionnaire following the guided experimental session, including questions about agency, ownership, and co-location of the real and virtual hand. The results of the questionnaire revealed a surprising effect regarding the subjective awareness as to the mismatch between real and virtual hand position (when present); In the robot-guided condition, participants largely did not perceive the 25° mismatch between the location of the virtual hand and that of their own hand (mean score = 4.8/6). In order to further investigate this surprising effect, we assessed whether it was specific to guided-adaptation, or was present also when participants performed active adaptation. Thus, we compared the questionnaire data of the guided condition, to the response to the same questionnaire collected in a condition of free VR active movement, either implementing a similar shift as that of the robot-guided condition ($N = 16$; 'shift VR') or with no shift ($N = 10$; 'no shift VR'; see methods for details). Interestingly, while in the active 'shift VR' condition participants reported misalignment between the virtual and real hand, in both robot-guided adaptation and in the 'no shift VR' condition the location ratings were equally high ($p = 0.92$), and no discrepancy was reported ($p < 0.001$ between shift VR and each of the other conditions,

Tukey test following 1-way ANOVA; Fig. 6d; 'Co-Location'). Correspondingly, ownership was higher in the active 'no shift VR' condition as compared to the 'shift VR' condition ($p = 0.007$), but there was no significant difference between ownership rating for the 'no shift VR' condition and robot-guided condition ($p = 0.28$). As expected, subjects did report reduced feeling of agency (control over the virtual hand) in the condition where the robot was guiding their movement compared to the 'no shift VR' condition ($p = 0.047$).

Taken together, the results of the guided adaptation condition show that although participants did not perform the reaching task actively, they still exhibited motor aftereffects in open-loop reaching, which did not generalize to reaching straight ahead task. However, unlike following an active adaptation condition, they remained unaware of the shift between the proprioceptive cues from their real hand and visual input coming from the virtual hand.

4. Discussion

We have implemented a novel procedure of PA, based on haptic-robotics and immersive virtual-reality. It allowed us to introduce a systematic shift of the visual feedback related to hand movements during a reaching task. Exploiting the advantages of VR environment, we were able to test the adaptation dynamics and subsequent aftereffects in both near and far space, in a gamified and ecological environment. Importantly, the combination of VR and a haptic robot enabled us to create also a condition of robotically-guided adaptation without active motor control. The present results show that our robotic VR-PA system successfully reproduces, in healthy participants, online adaptation during the training, as well as the PA aftereffects in subsequent motor tasks. For active adaptation conditions, these aftereffects were present both in an open loop-task and independently in a straight-ahead reaching task in absence of visual information, while they were not evident in a purely perceptual task. Comparable motor after-effects were present after adaptation in near and far space and generalized both the trained and non-trained sectors of space. Interestingly, similar motor aftereffects (in the open-loop task) were present also when the adaptation was performed via robotically-guided movements with no intentional motor control. Surprisingly, in the robotically-guided condition, participants were not aware of the mismatch between the position of

their real hand, as signaled by proprioception, and that of the virtual hand, as signaled by vision. We will discuss these results in light of previous motor adaptation findings, their contribution to the understanding of the adaptation process and possible applications for neglect rehabilitation.

4.1. Generalization across different sectors of space and different target positions

The new setup has allowed us also to test whether participants were equally able to adapt to the visual deviation when the target stimuli were placed either in the space within their reach or in the far space. Our participants showed similar adaptation dynamics when VR-PA training was performed in the near and in the far space (though reaching for far stimuli showed more trial-by-trial variability). We also tested the generalization of the training effects to the portion of space not directly submitted to the sensory-motor adaptation. Interestingly, the open-loop aftereffect was not space specific, such that training either in the near space or in the far space generated aftereffects both in the trained and in the non-trained sector of space. Previous studies on visuomotor rotation, in which participants had to reach towards targets while visual input of their hand trajectory was rotated using a computer screen, showed that training at a certain eccentricity induces aftereffects even when performing open-loop tests in an untrained eccentricity (Krakauer et al., 2000; Shabbott and Sainburg, 2010). However, the measured and trained portions of space were both within the peripersonal space, therefore these studies did not test the generalization of sensorimotor adaptation during reaching to the far, extrapersonal space.

In a recent study by Fleury et al. (2020), participants were trained on a throwing task and a pointing task while wearing prismatic glasses, and open-loop aftereffects were tested in both tasks. They found that true adaptation, with generalization of aftereffects to the untrained task, depends on the nature of the task performed (only pointing generalized to the throwing and not vice-versa), as well as on the level of mastery within the task (expert throwers with excellent throwing performance did show aftereffects that generalized to pointing task following prism exposure during a throwing task). In our study, although the VR gamification of the far task resembled a throwing game, the experimental setup of the near and far training and open-loop testing were made as similar as possible, to isolate the space component as the only essential difference between these conditions. Thus, the required action kinematic in the VR-PA task in the near and the far space is essentially the same (except the final part of releasing to shot), in order to facilitate the cross-over between the two tasks and the two portions of space. The fact that indeed we found a transfer of aftereffects from far training to the near space, implies that our far task was probably more similar to pointing than to throwing (as described in Fleury et al., 2020), and that participants reached a level of performance in which they sufficiently master the task.

Additionally, the experimental setup of the near and far training and open-loop testing were made as similar as possible, to isolate the space component as the only essential difference between these conditions. Thus, the motor execution in the VR-PA task in the near and the far space is essentially similar, though conceptually different (throwing versus reaching), facilitating the cross-over between the two tasks and the two portions of space. Our results extend previous findings by demonstrating that, when most other experimental properties are equalized (e.g., type of motor action, visual stimuli, task, reward etc.), the peripersonal and extrapersonal space training produce comparable, non-space-specific, aftereffects.

4.2. Generalization of motor aftereffects to a passive robot-guided condition

It is still not fully elucidated whether active motor control is required for achieving adaptation, or, by contrast, a visual-proprioceptive

mismatch is sufficient. There are evidence that merely imagined movements while wearing a prism with the absence of actual motor execution are sufficient for obtaining small, PA-like, aftereffects (Michel et al., 2013). In order to remove the active motor component from the adaptation process, while keeping all other adaptation parameters constant, we replayed example kinematics of an active adaptation session to implement the reaching movements of the participants in the robot-guided VR-PA condition. We found that following robot-guided VR-PA, open-loop aftereffects were similar to those that resulted from the active VR-PA, but with more variable non-significant effect in reaching straight ahead task. This implies that the intention to move the hand in combination with visual-proprioceptive disparity was enough to drive aftereffects in the absence of actual motor commands and motor error signal. In a previous study, Cressman and Henriques (2010) used a robotic device to passively train healthy participants in a visuomotor rotation paradigm with a cursor on a screen representing hand position. They found that performing the visuomotor adaptation either actively or passively produced comparable proprioceptive and motor aftereffects. Our results extend those previous findings by showing that sensorimotor adaptation under guided condition can be obtained also in an immersive VR set up, providing richer visual feedback. In addition, here we used the natural non-constrained movement kinematics of another person to guide participants' reaching movement during adaptation, rather than induced a movement along a linear path. Finally, in order to reduce participants' awareness of the manipulation, Cressman & Henriques introduced the visuo-motor rotation gradually over 40 trials, whereas in our study, the shift was introduced abruptly after 10 unshifted movements, as in the active condition. Surprisingly, even though when freely moving their hand subjects immediately perceived the shift between the real and the virtual hand, the participants in the guided condition seemed unaware of the mismatch between vision and proprioception during adaptation. It is possible that during the robot-guided condition, the absence of motor commands reduces the precision of the proprioceptive system in localizing one's own hand, thus hindering the perception of the shift of the visual hand. A recent study by Fang et al. (2019) presented a model for the extent of integration versus separation of visual and proprioceptive signals in a visuomotor rotation task. Monkeys performed a reaching task, while a virtual arm was shown at different levels of displacement with respect to the real arm. They found differential neuronal activation in the premotor cortex as a function of visuo-proprioceptive displacement, which discriminated whether the visual and proprioceptive cues from the hand were perceptually integrated into a unified hand position or not. We might speculate that in the robotically-guided condition, activation in the premotor areas, integrating or separating visual and proprioceptive inputs, is less consistent, which leads to reduced discriminability, and thus reduces awareness about the experimentally induced mismatch. Interestingly in Fang et al.'s (2019) study, in human participants the effect induced by the visually shifted virtual hand on reaching movements correlated with subjective ratings of ownership for the virtual hand. Accordingly, in our data, participants reported high ratings of ownership for the virtual hand in the robot-guided condition, indicating a stronger integration of visual and proprioceptive cues in that condition.

A recent study found that visuomotor adaptation in immersive VR relies more on cognitive strategies and explicit learning compared to an identical adaptation procedure on a computer screen (Anglin et al., 2017). However, the presence of an aftereffect in a condition where there was no awareness of the manipulation, implies a plastic recalibration of participants' sensory-motor reference frames, which should be mainly the result of implicit learning during the training.

4.3. Generalization of the training to a motor task without visual stimuli

The aftereffects we observed were not task or environment specific, but, at least for active VR-PA, were detectable in both open-loop task and reaching straight ahead task in complete darkness. Even without

any external stimulus, while participants performed a movement based only on proprioception and internal perception of space, a leftward bias was found following the active VR-PA training. This result was important to exclude that the aftereffect demonstrated in the open-loop task was not merely due to a contextual strategic leftward correction of the perceived VR environment acquired during the VR training, but represents a context-independent aftereffect (Fleury et al., 2019).

This finding is compatible with the effects shown using classic PA (Rossetti et al., 1998; Pisella et al., 2002; Sarri et al., 2008), as well as with works of Cressman & Henriques, which found a proprioceptive bias in perceived hand position following visuomotor adaptation (Cressman and Henriques, 2009; Henriques and Cressman, 2012). In neglect patients, pointing straight ahead is one of the most sensitive methods to detect a distortion of spatial reference frames (Sarri et al., 2008), as a rightward bias in the pointing straight ahead is present in almost all neglect patients. This task has been extensively used to assess the center of the reference frame for the representation of the external space, as indexed by proprioceptive coordinates (subjective bodily midline; Heilman et al., 1983; Karnath et al., 1991; Karnath, 1994; Richard et al., 2004). In addition, the correction towards the left of the subjective bodily midline is considered a key mechanism underlying neglect amelioration, and it is the therapeutic target of other manipulations, such as (trunk rotation, optokinetic nystagmus, caloric vestibular stimulation, transcutaneous electrical stimulation; Rubens, 1985; Karnath, 1994; Vallar et al., 1995; Wiart et al., 1997; Pizzamiglio et al., 2004). Thus, the finding of a deviation reaching straight ahead demonstrated here suggests that this novel VR-PA should be as effective as standard PA for neglect. In fact, participants performed only one straight ahead movement at the end of the session, instead of several movements as reported in other studies (e.g., Sarri et al., 2008). Consequently, data was variable, but still significant, which could imply that the real underlying effect is actually even more reliable than the one that we report here. In the guided condition, however, we could not confirm the same consistent change in the subjective straight ahead reaching, but instead results showed high variability. This variability could be due to methodological reasons: the fact that only one straight ahead movement was recorded in each session, and the fact that this active movement came immediately at the end of the guided VR-PA session, where participants were asked to remain passive. The lack of motor-proprioceptive aftereffect in the guided condition could be due either to the variability in our data, or suggest that the manipulation in the robotically-guided condition in normally able individuals is not consistent enough to induce a reliable effect at this level.

4.4. Rightward VR-PA training did not cross-over to a perceptual task

In our study, we did not find any influence of the VR-PA training on perceptual judgments in the landmark task. This finding might be at odds with neglect literature showing a significant improvement of spatial perception following PA treatment (Rossi et al., 1990; Pisella et al., 2002). However, the lateral biases visible in perceptual tasks in healthy participants as a result of PA usually appear only following leftward and not rightward PA. Indeed, several studies showed that the aftereffects in the landmark task in healthy participants are limited, as most studies did not find consistent perceptual aftereffects following rightward-shifting prisms (Colent et al., 2000; Berberovic and Mattingley, 2003; Bultitude and Rafal, 2010; Schintu et al., 2014, 2017). In light of these results, we acknowledge the limitation of our choice to perform a perceptual task following rightward rather than leftward adaptation training, which was done in order to serve as a basis for future application for neglect rehabilitation (which is usually left hemispatial neglect; Cubelli, 2017). We predict that a similar paradigm with a leftward shift will show symmetrical visuo-motor and motor-proprioceptive aftereffects in healthy subjects, but might be more efficient in inducing perceptual aftereffects due to the inherent spatial biases reported for healthy participants ('pseudoneglect'; Bowers and Heilman, 1980), and

reports in previous literature using leftward shift virtual visuomotor shift in healthy participants (Numao et al., 2019).

4.5. Theoretical implications on possible neural correlates of adaptation

Our adaptation VR experiments revealed a stable motor aftereffect following training in different sectors of space, and even in training without active motor control. The stability of the aftereffect implies that the spatial realignment induced by the adaptation training is not stimulus-specific, but affects multiple spatial representations involved in motor tasks. Recent imaging studies demonstrate that following prism adaptation, spontaneous brain activity is modulated in heteromodal networks that are involved in attention or in intrinsic, self-related processes (Schintu et al., 2019; Tsujimoto et al., 2019; Wilf et al., 2019). In neglect patients, prism adaptation modulates subsequent cortical responses in attentional and default mode regions, but not in visual or motor regions (Saj et al., 2013; Crottaz-Herbette et al., 2017). Moreover, a recent study showed that patient responsiveness to PA relies on temporo-parietal and prefrontal pathways, which further strengthens the notion that sensory-independent networks stand in the basis of the adaptation (Lunven et al., 2019).

Our behavioural results are coherent with those imaging studies in suggesting that adaptation aftereffects do not rely on specific low-level features of the sensory inputs or motor responses. However, sensory impairments (e.g., hemianopia, Serino et al., 2006), attentional impairments for specific sectors of space (e.g., near vs. far space neglect, Halligan and Marshall, 1991) or motor impairment in patients might hinder adaptation and thus reduce neglect recovery. Thus, it might be useful to adapt the adaptation protocol to the patients' impairments, as the current robotic-VR system allows.

There is one important theoretical point to address. There is evidence that visuo-motor rotation, among other classes of dynamic perturbations such as force fields, is a different process than prismatic adaptation. In a recent review, Fleury et al. (2019) suggested that there is a fundamental difference between prism adaptation and visuomotor rotation, namely, the difference in credit attribution. While in prism adaptation the ownership of the perceived hand is unquestionable, in visuomotor rotation with an external apparatus, the perturbation might be attributed to the interface properties, rather than to the adaptation of the subject's own sensorimotor system. Nevertheless, our setup presented adaptation training in a 1st person perspective, with a virtual hand (at least in the practice session), and our data demonstrate high embodiment of the virtual effector (see Fig. 6d). Thus, it is unlikely that participants attributed the spatial discrepancy only to the external interface and used strategic mechanisms to compensate for it.

4.6. Applications for neglect rehabilitation

These results might have interesting implications on neglect treatment. First, there are cases in which neglect selectively affects the peripersonal or the extrapersonal space (Bisiach et al., 1986; Halligan and Marshall, 1991), with distinct lesion locations for the different subtypes (Committeri et al., 2007). Since our setup enables adaptation in the near or the far space, it will be interesting to test whether these patients, unlike healthy participants, will show differential aftereffects depending on the trained portion of space, that will eventually be applied for designing an appropriate training protocol. Previous research in non-VR environments have shown that the beneficial effects of PA performed in the near space generalize to neglect tasks with stimuli presented in the extrapersonal space (Serino et al., 2006; Patané et al., 2016). However, to the best of our knowledge, there is no conclusive evidence regarding effectiveness of training neglect patients in the far space, nor a systematic assessment of the near/far visuo-motor aftereffects as induced by near/far training.

Second, our setup enables robot-guided assisted adaptation training. Today, neglect rehabilitation via PA is performed with the ipsilesional

limb. However, patients suffering from hemiplegia could have a double benefit from assisted-PA training, meaning that the patient can even be trained by using the contralesional hand, which is usually hemiparetic or hemiplegic. This could offer the intriguing possibility of combining training for the visuo-spatial components, as well as for the motor component of neglect and even upper limb motor rehabilitation. Moreover, to accommodate for individual patient differences in range of motion, there is a big advantage in using the patient's own kinematics from the healthy limb to guide training of the hemiplegic arm.

Third, the VR adaptation setup enables to create larger deviations than those applicable in standard prismatic lenses, which are limited by optical and material properties. In healthy participants, larger deviations have been shown to induce bigger motor-proprioceptive and cognitive effects (Michel, 2015; Striemer et al., 2016; Gammeri et al., 2018). Therefore, VR-PA has the potential of creating more effective aftereffect, thus potentially being more effective for neglect rehabilitation (Facchin et al., 2013). Indeed, our manipulation with 25° shift resulted in an average of 6.1° open-loop aftereffect, which represents around 25% of the shift, in accordance with 30% reported in PA literature, meaning between 3° and 4° for the 10° or 15° optical deviations normally induced with prismatic lenses (Facchin et al., 2013; Michel and Cruz, 2015; Striemer et al., 2016).

Forth, healthy participants might rely more on their intact vision when performing sensory-motor tasks, but neglect patients or patients with hemianopia have impaired ability to use their visual inputs (Serino et al., 2006). In these cases, haptic feedback that is properly adjusted for the individual patient might help achieve more efficient rehabilitation. Forth, the flexibility of the VR-robotics setup can allow adjustment of the level of difficulty (e.g. size and the location of the target, the amount of shift) and the amount of feedback (proximal or distal feedback regarding hand position) can be adjusted to fit the impairments and capabilities of each patient individually (Ladavas et al., 2011).

And finally, the interesting gamified environment, along with multisensory feedback on motor performance, can enhance motivation to possibly enable longer-term adaptation training, which have been shown to benefit patient rehabilitation (Frassinetti et al., 2002; Serino et al., 2009).

5. Conclusion

Using novel setup of robotic virtual reality-based prism adaptation training, we found that motor adaptation aftereffects are invariant to the specific parameters of the adaptation training. VR-PA can induce aftereffects that generalize to different tasks and contexts, although not to a purely perceptual task. Adaptation in either the near or the far space induces aftereffects also in the untrained portion of space, and even robot-guided passive adaptation training without active motor control can induce subsequent motor aftereffects. The ecological and immersive setup, along with the haptic robotic device is a promising tool to treat neglect patients, with and without motor impairment.

CRedit author statement

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Appendix A. Supplementary data

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References

- Anglin, J., Sugiyama, T., Liew, S.-L., 2017. Visuomotor adaptation in head-mounted virtual reality versus conventional training. *Sci. Rep.* 7, 1–8.
- Barrett, A., Goedert, K.M., Basso, J.C., 2012. Prism adaptation for spatial neglect after stroke: translational practice gaps. *Nat. Rev. Neurol.* 8, 567.
- Bassolino, M., Franza, M., Bello Ruiz, J., Pinardi, M., Schmidlin, T., Stephan, M., Solcà, M., Serino, A., Blanke, O., 2018. Non-invasive brain stimulation of motor cortex induces embodiment when integrated with virtual reality feedback. *Eur. J. Neurosci.* 47, 790–799.
- Bedford, F.L., 1999. Keeping perception accurate. *Trends Cognit. Sci.* 3, 4–11.
- Berberovic, N., Mattingley, J.B., 2003. Effects of prismatic adaptation on judgements of spatial extent in peripersonal and extrapersonal space. *Neuropsychologia* 41, 493–503.
- Bisiach, E., Perani, D., Vallar, G., Berti, A., 1986. Unilateral neglect: personal and extrapersonal. *Neuropsychologia* 24, 759–767.
- Bowers, D., Heilman, K.M., 1980. Pseudoneglect: effects of hemispace on a tactile line bisection task. *Neuropsychologia* 18, 491–498.
- Bultitude, J.H., Rafal, R.D., 2010. Amelioration of right spatial neglect after visuo-motor adaptation to leftward-shifting prisms. *Cortex* 46, 404–406.
- Carter, A., Foreman, M., Martin, C., Fitterer, S., Pioppo, A., Connor, L., Engsborg, J., 2016. Inducing visuomotor adaptation using virtual reality gaming with a virtual shift as a treatment for unilateral spatial neglect. *Journal of Intellectual Disability: Diagnosis and Treatment* 4, 170–184.
- Chamond, A.S., Frank, R.C., Taylor, K., Eskes, G.A., 2018. The effects of prism adaptation on daily life activities in patients with visuospatial neglect: a systematic review. *Neuropsychol. Rehabil.* 28, 491–514.
- Choi, Y.S., Lee, K.W., Lee, J.H., Kim, S.B., Park, G.T., Lee, S.J., 2016. The effect of an upper limb rehabilitation robot on hemispatial neglect in stroke patients. *Annals of rehabilitation medicine* 40, 611–619.
- Colent, C., Pisella, L., Bernieri, C., Rode, G., Rossetti, Y., 2000. Cognitive bias induced by visuo-motor adaptation to prisms: a simulation of unilateral neglect in normal individuals? *Neuroreport* 11, 1899–1902.
- Committeri, G., Pitzalis, S., Galati, G., Patria, F., Pelle, G., Sabatini, U., Castriota-Scanderbeg, A., Piccardi, L., Guariglia, C., Pizzamiglio, L., 2007. Neural bases of personal and extrapersonal neglect in humans. *Brain* 130, 431–441.
- Cressman, E.K., Henriques, D.Y., 2009. Sensory recalibration of hand position following visuomotor adaptation. *J. Neurophysiol.* 102, 3505–3518.
- Cressman, E.K., Henriques, D.Y., 2010. Reach adaptation and proprioceptive recalibration following exposure to misaligned sensory input. *J. Neurophysiol.* 103, 1888–1895.
- Crottaz-Herbette, S., Fornari, E., Notter, M.P., Bindschaedler, C., Manzoni, L., Clarke, S., 2017. Reshaping the brain after stroke: the effect of prismatic adaptation in patients with right brain damage. *Neuropsychologia* 104, 54–63.
- Cubelli, R., 2017. Definition: spatial neglect. *Cortex. A Journal Devoted to the Study of the Nervous System and Behavior*.
- Facchin, A., Beschin, N., Toraldo, A., Cisari, C., Daini, R., 2013. Aftereffect induced by prisms of different power in the rehabilitation of neglect: a multiple single case report. *NeuroRehabilitation* 32, 839–853.
- Facchin, A., Bultitude, J.H., Mornati, G., Peverelli, M., Daini, R., 2020. A comparison of prism adaptation with terminal versus concurrent exposure on sensorimotor changes and spatial neglect. *Neuropsychol. Rehabil.* 30, 613–640.
- Fang, W., Li, J., Qi, G., Li, S., Sigman, M., Wang, L., 2019. Statistical inference of body representation in the macaque brain. *Proc. Natl. Acad. Sci. Unit. States Am.* 116, 20151–20157.
- Fleury, L., Pastor, D., Revol, P., Delporte, L., Rossetti, Y., 2020. Inter-task transfer of prism adaptation depends on exposed task mastery. *Sci. Rep.* 10, 1–16.
- Fleury, L., Prablanc, C., Priot, A.-E., 2019. Do prism and other adaptation paradigms really measure the same processes? *Cortex* 119, 480–496.
- Frassinetti, F., Angeli, V., Meneghello, F., Avanzi, S., Ladavas, E., 2002. Long-lasting amelioration of visuospatial neglect by prism adaptation. *Brain* 125, 608–623.
- Gammeri, R., Turri, F., Ricci, R., Ptak, R., 2018. Adaptation to virtual prisms and its relevance for neglect rehabilitation: a single-blind dose-response study with healthy participants. *Neuropsychol. Rehabil.* 1–14.
- Halligan, P.W., Marshall, J.C., 1991. Left neglect for near but not far space in man. *Nature* 350, 498.
- Harris, C.S., 1963. Adaptation to displaced vision: visual, motor, or proprioceptive change? *Science* 140, 812–813.
- Heilman, K.M., Bowers, D., Watson, R.T., 1983. Performance on hemispatial pointing task by patients with neglect syndrome. *Neurology* 33, 661–661.
- Henriques, D.Y., Cressman, E.K., 2012. Visuomotor adaptation and proprioceptive recalibration. *J. Mot. Behav.* 44, 435–444.

- Karnath, H., Schenkel, P., Fischer, B., 1991. Trunk orientation as the determining factor of the 'contralateral' deficit in the neglect syndrome and as the physical anchor of the internal representation of body orientation in space. *Brain* 114, 1997–2014.
- Karnath, H.-O., 1994. Subjective body orientation in neglect and the interactive contribution of neck muscle proprioception and vestibular stimulation. *Brain* 117, 1001–1012.
- Katz, N., Ring, H., Naveh, Y., Kizony, R., Feintuch, U., Weiss, P., 2005. Interactive virtual environment training for safe street crossing of right hemisphere stroke patients with unilateral spatial neglect. *Disabil. Rehabil.* 27, 1235–1244.
- Kim, Y.M., Chun, M.H., Yun, G.J., Song, Y.J., Young, H.E., 2011. The effect of virtual reality training on unilateral spatial neglect in stroke patients. *Annals of rehabilitation medicine* 35, 309–315.
- Kluzik, J., Diedrichsen, J., Shadmehr, R., Bastian, A.J., 2008. Reach adaptation: what determines whether we learn an internal model of the tool or adapt the model of our arm? *J. Neurophysiol.* 100, 1455–1464.
- Krakauer, J.W., Pine, Z.M., Ghilardi, M.-F., Ghez, C., 2000. Learning of visuomotor transformations for vectorial planning of reaching trajectories. *J. Neurosci.* 20, 8916–8924.
- Ladavas, E., Bonifazi, S., Catena, L., Serino, A., 2011. Neglect rehabilitation by prism adaptation: different procedures have different impacts. *Neuropsychologia* 49, 1136–1145.
- Lunven, M., Rode, G., Boulon, C., Duret, C., Migliaccio, R., Chevrillon, E., de Schotten, M.T., Bartolomeo, P., 2019. Anatomical predictors of successful prism adaptation in chronic visual neglect. *Cortex* 120, 629–641.
- Makin, T.R., Wilf, M., Schwartz, I., Zohary, E., 2010. Amputees "neglect" the space near their missing hand. *Psychol. Sci.* 21, 55–57.
- McIntosh, R.D., Brown, B.M., Young, L., 2019. Meta-analysis of the visuospatial aftereffects of prism adaptation, with two novel experiments. *Cortex* 111, 256–273.
- Michel, C., 2015. Beyond the sensorimotor plasticity: cognitive expansion of prism adaptation in healthy individuals. *Front. Psychol.* 6, 1979.
- Michel, C., Cruz, R., 2015. Prism adaptation power on spatial cognition: adaptation to different optical deviations in healthy individuals. *Neurosci. Lett.* 590, 145–149.
- Michel, C., Gaveau, J., Pozzo, T., Papaxanthis, C., 2013. Prism adaptation by mental practice. *Cortex* 49, 2249–2259.
- Michel, C., Pisella, L., Halligan, P.W., Luauté, J., Rode, G., Boisson, D., Rossetti, Y., 2003. Simulating unilateral neglect in normals using prism adaptation: implications for theory. *Neuropsychologia* 41, 25–39.
- Numao, T., Fujita, Y., Ichikawa, K., Ide, S., Katori, H., Shimada, T., Ota, K., Ikeda, Y., Amimoto, K., 2019. Leftward optical shift induces bias in line bisection: a study with healthy subjects using a head-mounted display. *Progress in Rehabilitation Medicine* 4, 20190008.
- Olejnik, S., Algina, J., 2003. Generalized eta and omega squared statistics: measures of effect size for some common research designs. *Psychol. Methods* 8, 434.
- Patané, I., Farné, A., Frassinetti, F., 2016. Prismatic Adaptation Induces Plastic Changes onto Spatial and Temporal Domains in Near and Far Space. *Neural plasticity*, 2016.
- Pisella, L., Rode, G., Farné, A., Boisson, D., Rossetti, Y., 2002. Dissociated long lasting improvements of straight-ahead pointing and line bisection tasks in two hemineglect patients. *Neuropsychologia* 40, 327–334.
- Pizzamiglio, L., Fasotti, L., Jehkonen, M., Antonucci, G., Magnotti, L., Boelen, D., Asa, S., 2004. The use of optokinetic stimulation in rehabilitation of the hemineglect disorder. *Cortex* 40, 441–450.
- Prablanc, C., Panico, F., Fleury, L., Pisella, L., Nijboer, T., Kitazawa, S., Rossetti, Y., 2020. Adapting terminology: clarifying prism adaptation vocabulary, concepts, and methods. *Neurosci. Res.* 153, 8–21.
- Ramos, A.A., Hörning, E.C., Wilms, I.L., 2019. Simulated prism exposure in immersed virtual reality produces larger prismatic after-effects than standard prism exposure in healthy subjects. *PLoS One* 14.
- Redding, G.M., Rossetti, Y., Wallace, B., 2005. Applications of prism adaptation: a tutorial in theory and method. *Neurosci. Biobehav. Rev.* 29, 431–444.
- Redding, G.M., Wallace, B., 1996. Adaptive spatial alignment and strategic perceptual-motor control. *J. Exp. Psychol. Hum. Percept. Perform.* 22, 379.
- Richard, C., Rousseaux, M., Saj, A., Honoré, J., 2004. Straight ahead in spatial neglect: Evidence that space is shifted, not rotated. *Neurology* 63, 2136–2138.
- Rossetti, Y., Rode, G., Pisella, L., Farné, A., Li, L., Boisson, D., Perenin, M.-T., 1998. Prism adaptation to a rightward optical deviation rehabilitates left hemispatial neglect. *Nature* 395, 166.
- Rossi, P.W., Kheifets, S., Reding, M.J., 1990. Fresnel prisms improve visual perception in stroke patients with homonymous hemianopia or unilateral visual neglect. *Neurology* 40, 1597–1597.
- Rubens, A.B., 1985. Caloric stimulation and unilateral visual neglect. *Neurology* 35, 1019–1019.
- Saj, A., Cojan, Y., Vocat, R., Luauté, J., Vuilleumier, P., 2013. Prism adaptation enhances activity of intact fronto-parietal areas in both hemispheres in neglect patients. *Cortex* 49, 107–119.
- Sarri, M., Greenwood, R., Kalra, L., Papps, B., Husain, M., Driver, J., 2008. Prism adaptation aftereffects in stroke patients with spatial neglect: pathological effects on subjective straight ahead but not visual open-loop pointing. *Neuropsychologia* 46, 1069–1080.
- Schintu, S., Freedberg, M., Gotts, S.J., Cunningham, C.A., Alam, Z.M., Shomstein, S., Wassermann, E.M., 2019. Prism Adaptation Modulates Connectivity of the Intraparietal Sulcus with Multiple Brain Networks. *bioRxiv*, p. 683078.
- Schintu, S., Patané, I., Caldano, M., Salemmé, R., Reilly, K.T., Pisella, L., Farné, A., 2017. The asymmetrical effect of leftward and rightward prisms on intact visuospatial cognition. *Cortex* 97, 23–31.
- Schintu, S., Pisella, L., Jacobs, S., Salemmé, R., Reilly, K.T., Farné, A., 2014. Prism adaptation in the healthy brain: the shift in line bisection judgments is long lasting and fluctuates. *Neuropsychologia* 53, 165–170.
- Serino, A., Angeli, V., Frassinetti, F., Ladavas, E., 2006. Mechanisms underlying neglect recovery after prism adaptation. *Neuropsychologia* 44, 1068–1078.
- Serino, A., Barbisani, M., Rinaldesi, M.L., Ladavas, E., 2009. Effectiveness of prism adaptation in neglect rehabilitation: A controlled trial study. *Stroke* 40, 1392–1398.
- Shabbott, B.A., Sainburg, R.L., 2010. Learning a visuomotor rotation: simultaneous visual and proprioceptive information is crucial for visuomotor remapping. *Exp. Brain Res.* 203, 75–87.
- Striener, C.L., Russell, K., Nath, P., 2016. Prism adaptation magnitude has differential influences on perceptual versus manual responses. *Exp. Brain Res.* 234, 2761–2772.
- Tsirlin, I., Dupierri, E., Chokron, S., Coquillart, S., Ohlmann, T., 2009. Uses of virtual reality for diagnosis, rehabilitation and study of unilateral spatial neglect: review and analysis. *Cyberpsychol. Behav.* 12, 175–181.
- Tsujimoto, K., Mizuno, K., Nishida, D., Tahara, M., Yamada, E., Shindo, S., Kasuga, S., Liu, M., 2019. Prism adaptation changes resting-state functional connectivity in the dorsal stream of visual attention networks in healthy adults: a fMRI study. *Cortex* 119, 594–605.
- Vallar, G., Rusconi, M., Barozzi, S., Bernardini, B., Ovadia, D., Papagno, C., Cesarani, A., 1995. Improvement of left visuo-spatial hemineglect by left-sided transcutaneous electrical stimulation. *Neuropsychologia* 33, 73–82.
- Varalta, V., Picelli, A., Fonte, C., Montemazzi, G., La Marchina, E., Smania, N., 2014. Effects of contralesional robot-assisted hand training in patients with unilateral spatial neglect following stroke: a case series study. *J. NeuroEng. Rehabil.* 11, 160.
- Wiaart, L., Saint Côme, A.B., Debelleix, X., Petit, H., Joseph, P.A., Mazaux, J.M., Barat, M., 1997. Unilateral neglect syndrome rehabilitation by trunk rotation and scanning training. *Arch. Phys. Med. Rehabil.* 78, 424–429.
- Wilf, M., Serino, A., Clarke, S., Crotaz-Herbette, S., 2019. Prism adaptation enhances decoupling between the default mode network and the attentional networks. *Neuroimage* 200, 210–220.